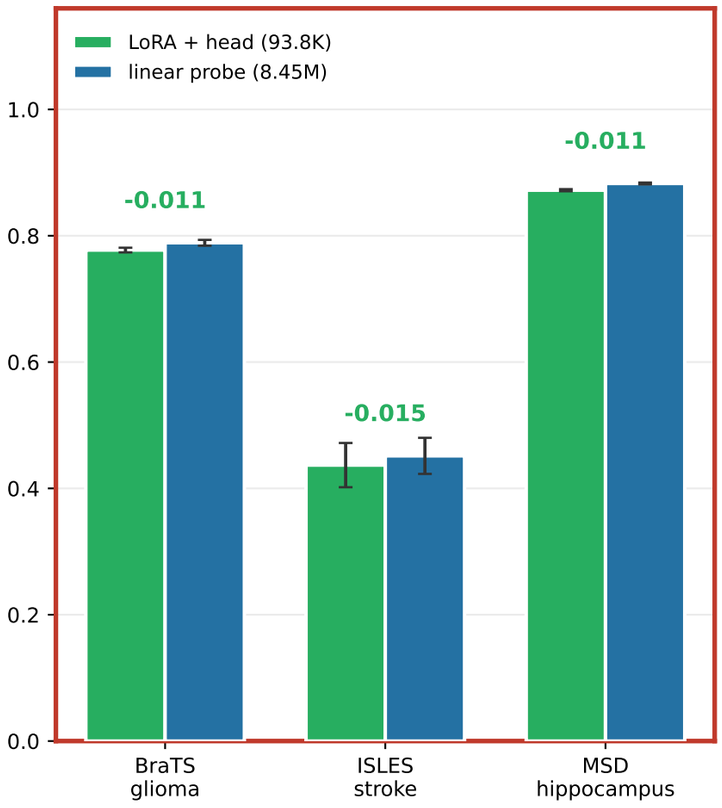
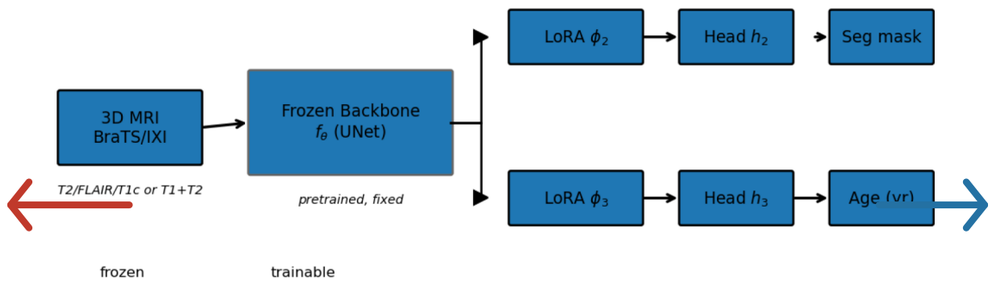
Few-Shot Continual Learning for 3D Brain MRI with Frozen Foundation Models

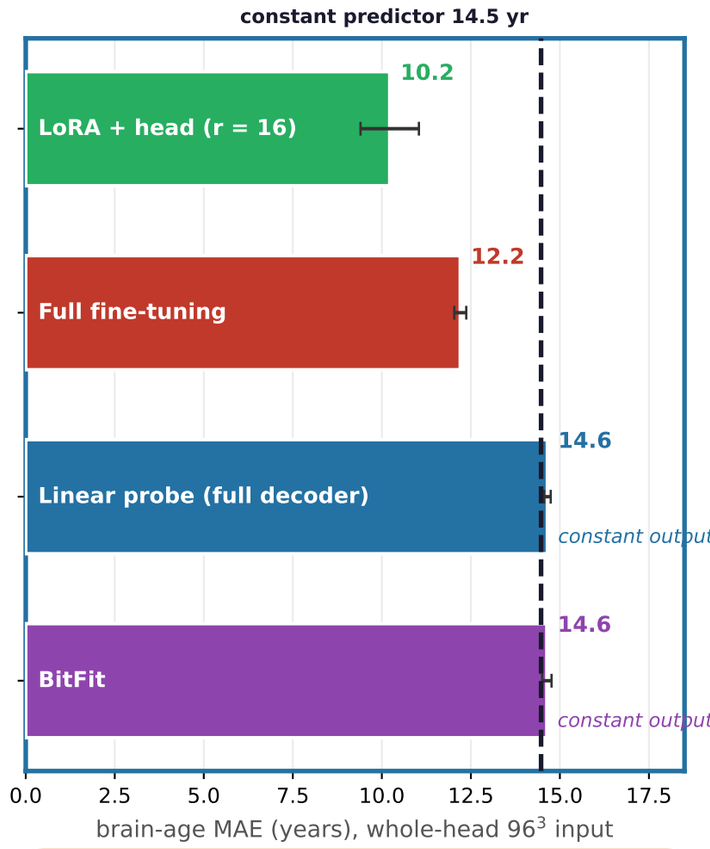
Segmentation: 90× cheaper, −0.01 Dice



Frozen backbone + per-task LoRA



Global tasks: only LoRA learns them



10.2 yr | 0.88 AUC

Two global tasks, same answer

Brain age 10.2 vs 14.5 constant; sex AUC 0.88 vs 0.82 full fine-tune; linear probe outputs a constant on both

0.011-0.015

Segmentation margin, 3 tasks

Same gap on glioma, stroke and hippocampus despite differing in channels, pathology and scale

AUC 0.72, acc 0.50

Signal present, unusable

Frozen features rank sex far above chance, yet a 1K head gives a constant decision — the bottleneck is geometry